A Study on the Application of XAI to Diffusion Models for Synthesizing Electrocardiograms

1. Introduction

Between 1990 and 2001, Regulatory authorities or pharmaceutical companies discontinued the development of numerous pharmaceutical agents due to the risk of Torsade de Pointes (TdP). Researchers have found that many drugs that induce TdP prolong the QT interval on electrocardiograms (ECGs) making ECG evaluation an essential component of modern clinical drug development. However, each ECG measurement costs approximately USD 50, and large-scale assessments incur substantial financial burdens. Furthermore, when missing data occur, re-measurement is required, resulting in additional costs. To address these issues, AI-based ECG synthesis and prediction have been reported as effective alternatives.

In previous studies, Researchers investigated ECG waveform synthesis using SSSD^{S4}. They designed this model for imputing missing data. The results demonstrated that diffusion models incorporating the Structured State Space Sequence (S4) layer, which is designed to capture long-range dependencies in time-series data, fit well for ECG synthesis. However, the generation process of AI-based models is generally considered a “black box,” and the decision-making mechanisms behind the predicted outputs remain opaque. As a result. It is often unclear on what basis the model generates ECG waveforms and it is difficult to rule out the possibility that the correct waveforms were generated by chance. Therefore, concerns regarding the reliability of the outputs persist.

To address this issue, the present study proposes enhancing the interpretability of AI-generated outputs by applying Explainable AI (XAI) techniques. Specifically, we apply XAI to the SSSDS4 generative model for ECG synthesis, aiming to visualize time-series features that are considered crucial during the generation process and thereby improve the reliability of the generated results.

2. Generative Models and Explainable AI

2.1 SSSD^{S4}

SSSD^{S4} is a generative model based on the diffusion model DiffWave, in which the diffusion layers within residual blockswere replaced by researchers with layers from the Structured State Space Sequence (S4) model. This modification allows the model to efficiently model of long-range dependencies in time-series data.

Researchers developed The S4 model is a method designed for long-range dependency modeling, based on the state space model framework. Equation (1) presents its mathematical formulation. This model maps a one-dimensional input u(t)u(t) to a latent state x(t)x(t), which is then transformed into the output signal y(t)y(t). The coefficient matrices A,B,C, and D are learnable parameters optimized via gradient descent.

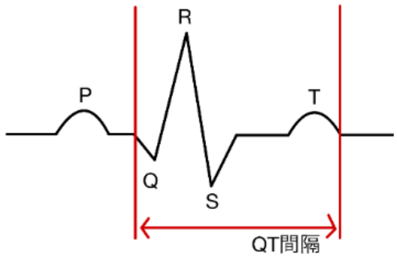

Figure 1. Example of a Typical Electrocardiogram (ECG) Waveform

$$x\,(t) = \boldsymbol{A}x(t) + \boldsymbol{B}u(t) \quad and \quad y(t) = \boldsymbol{C}x(t) + \boldsymbol{D}u(t) \qquad (1)$$

2.2 XAI

In this study, three model-agnostic XAI methods were employed: LIME, KernelSHAP—which is based on SHAP theory applied to LIME—and its optimized variant, FastSHAP. The contribution of each input feature to the model output is quantified by SHAP based on cooperative game theory, while feature importance is estimated by LIME through the generation of perturbations around a specific data point and the analysis of the corresponding model predictions.

3. Dataset and the Proposed Approach

3.1 Dataset

This study was conducted with the approval of the Ethics Committee of Jichi Medical University Saitama Medical Center (Approval No. Rin S23-071). For training and validation, they used neonatal ECG data that they acquired at a sampling rate of 1000 Hz in the NICU of the same center. They down sampled, standardized, and segmented the ECG voltage data from six neonates into 200-point (2-second) intervals to construct the dataset.

3.2 Proposed Approach

If one heartbeat can be predicted from approximately two preceding heartbeats, a significant contribution can be made to reducing the number of measurements required in clinical evaluation. Accordingly, the application of explainable AI (XAI) is proposed to a model that was trained to generate the final 60 points of input data—corresponding to approximately one heartbeat—artificially masked as missing. Since directly applying XAI methods to the waveform output of SSSDS4 is challenging, we extended the model by incorporating both a regression and a classification framework to convert waveform outputs into numerical values.

Specifically, R-peaks were detected from the generated waveforms using the Christov algorithm, and their positions were used as outputs in the regression model (Figure 1). For the classification model, each point in the waveform was treated as a class, and the regression output was converted into a pseudo-probability distribution by assuming a normal distribution with the predicted position as the mean and a standard deviation of 3 (Figure 2). KernelSHAP and FastSHAP were applied to the regression model, while LIME was used for the classification model to conduct the XAI analysis.

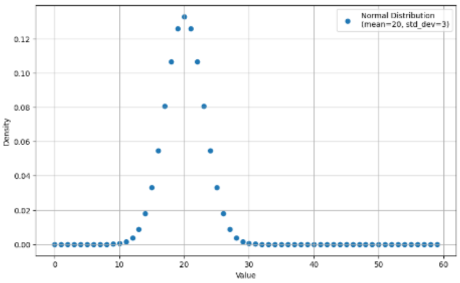

Figure 2. Example of Class Probabilities

4. Application of Explainable AI to SSSDS4

4.1 Application of SHAP to SSSDS4

SHAP represents the contribution of each input feature to the model output using SHAP values.A higher SHAP value indicates a greater influence on the output, reflecting the features the model considers important. Figures 3 and 4 show the results of applying KernelSHAP and FastSHAP to SSSDS4, respectively. The parameters were adjusted so that both methods required approximately 1.5 days of computation time. In the figures, red indicates the input, green the synthesized ECG waveform, and blue bars represent the absolute SHAP values at each point.

Comparison showed that KernelSHAP failed to output SHAP values at many points (Figure 3), whereas FastSHAP successfully computed values at more points (Figure 4). Therefore, under equal computational time, FastSHAP proved more effective for interpreting SSSDS4 and was adopted in this study.

An example result of applying FastSHAP to multiple data samples is shown in Figure 5. High SHAP values were frequently observed near the generated segment and around the R and T wave peaks. Additionally, SHAP values were also present at earlier time points (near point 0), suggesting that SSSDS4 emphasizes not only regions near the target output and key waveform components, but also captures long-term dependencies from past data.

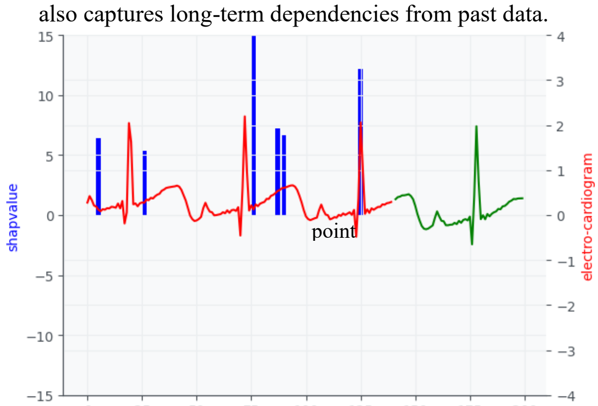

Figure 3. SHAP Values Computed Using KernelSHAP

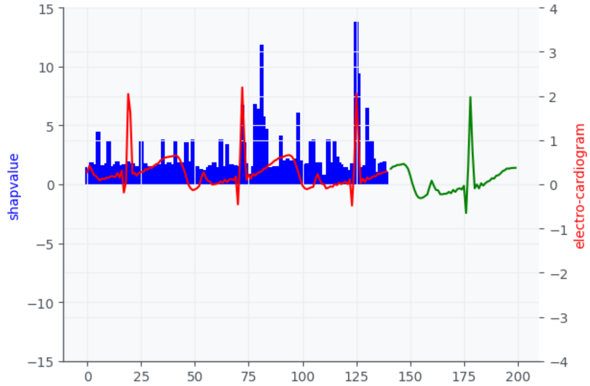

Figure 4. SHAP Values Computed Using FastSHAP

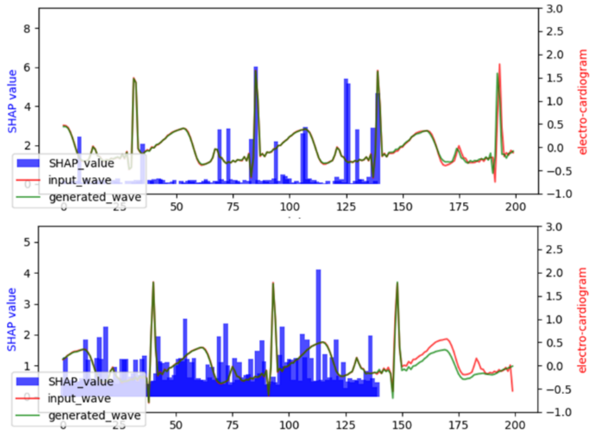

Figure 5. Example of SHAP Values Computed Using FastSHAP

4.2 Application of LIME to SSSDS4

To apply LIME to time-series data, the input was divided into 70 segments and perturbed individually. Figure 6 highlights the five most influential segments in yellow, which include the peaks of the R and T waves. Similar patterns were observed with SHAP, suggesting that SSSDS4 emphasizes the periodicity of the R and T waves.

5. Conclusion

In this study, we proposed an XAI-based method to interpret the ECG generative model SSSDS4 by visualizing the features the model prioritizes. The application of XAI suggested that SSSDS4 focuses not only on regions near the generated segment but also on past information and waveform periodicity. Future work includes evaluating feature importance across more data and verifying the reliability of the results from medical and pharmaceutical perspectives.

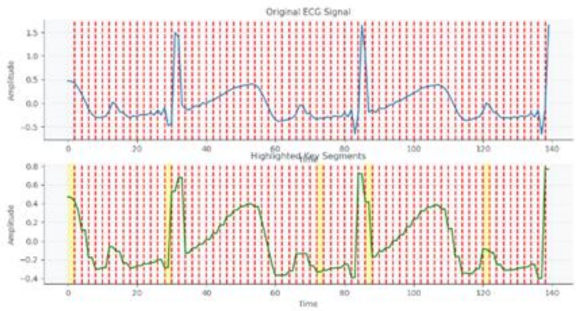

Figure 6. Results of Applying LIME

使用した AI

Chat GPT 4o , Copilot の 2 つを用いて Introduction をお互いで評価した結果 Chat GPT 4o の方が優れているように感じたため、英訳には基本的に Chat GPT 4o を使用した

生成した手法

初めに日本語の状態で Chat GPT を使い最小化を行った。その際に日本語でおかしな部分がないかを確認して、意味合いが違う部分は修正した

例

次に、最小化した日本語を英訳した

英語の読解力に自身がないため、英訳したチャットとは別のチャットで英訳したものを日本語に戻した。その際には DeepL なども使用した。そしておかしな部分がないかを確認した。

最後に、AI なども使用しつつ、受動態になっている部分の能動態への変更などを行った。